Probabilistic Machine Learning

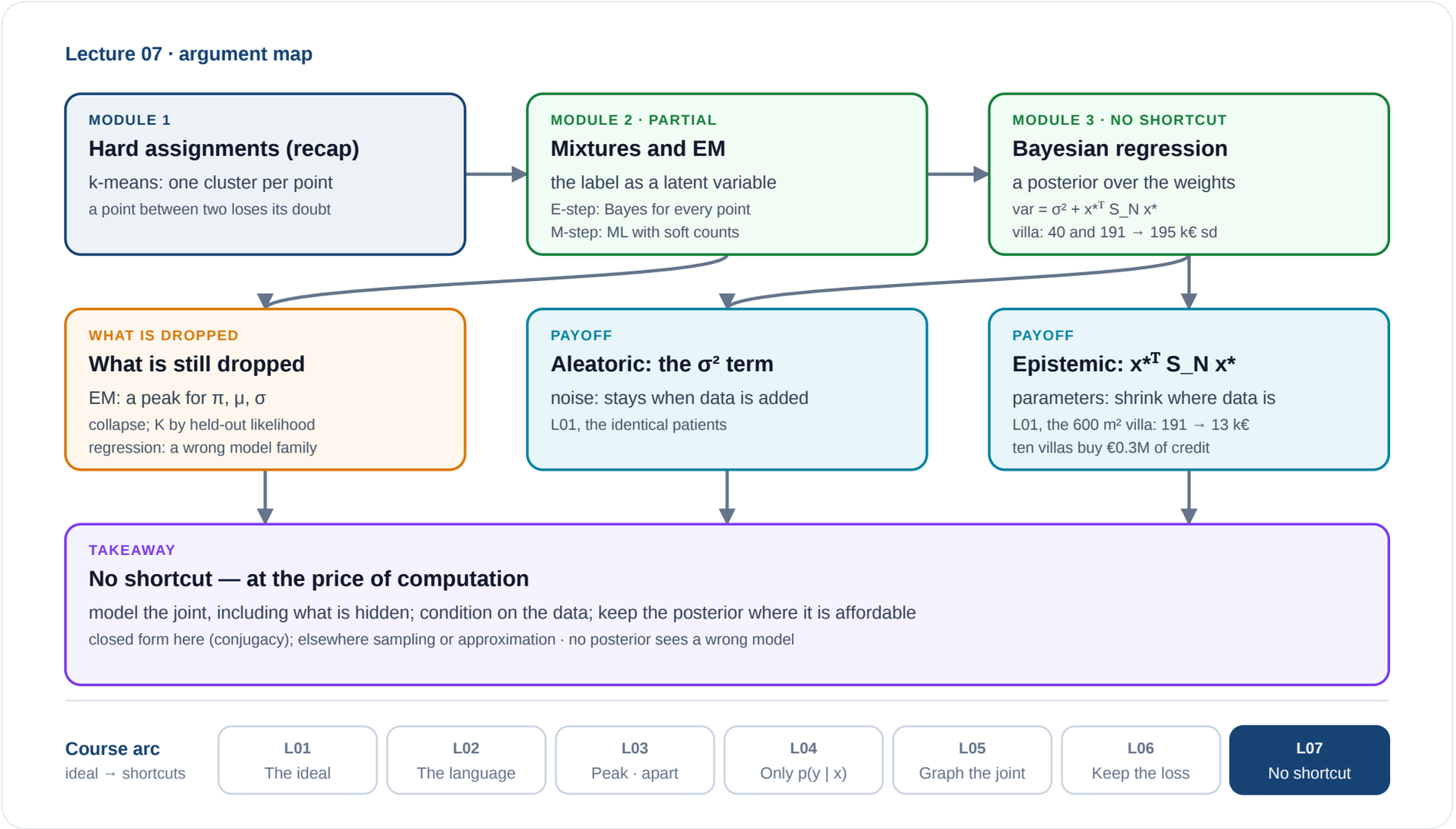
Lecture 07: Selected Probabilistic Algorithms

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Speaker notes

Reworked 2026-09-26 (docs/reviews/PML-L05-L07-REVIEW-2026-09.md §5.3). The closing lecture carries out Lecture 01's ideal — model the joint, condition on the data, read off answers — twice, and says honestly how far each gets. Module 1 is a three-slide k-means recap (the students know clustering) that ends on the question k-means cannot answer. Module 2: Gaussian mixtures and EM, a *partial* shortcut — a posterior over the hidden labels, a peak for the parameters. Module 3: Bayesian regression on Lecture 01's house prices, with the predictive variance split into a noise and a parameter term, in numbers — the promise of Lecture 01. Module 4 closes the course arc.

Argument Map



Speaker notes

The end of the arc. k-means gives every point one label; a mixture makes the label a latent variable and EM conditions on it — but keeps only a peak for the parameters, so EM is a partial shortcut. Bayesian regression keeps the posterior over the weights, and its predictive variance splits into the noise term (aleatoric — Lecture 01's identical patients) and the parameter term (epistemic — Lecture 01's 600 m² villa). The takeaway row says why every other lecture took a shortcut: the ideal costs computation, and even it cannot see a wrong model family.

Module 1

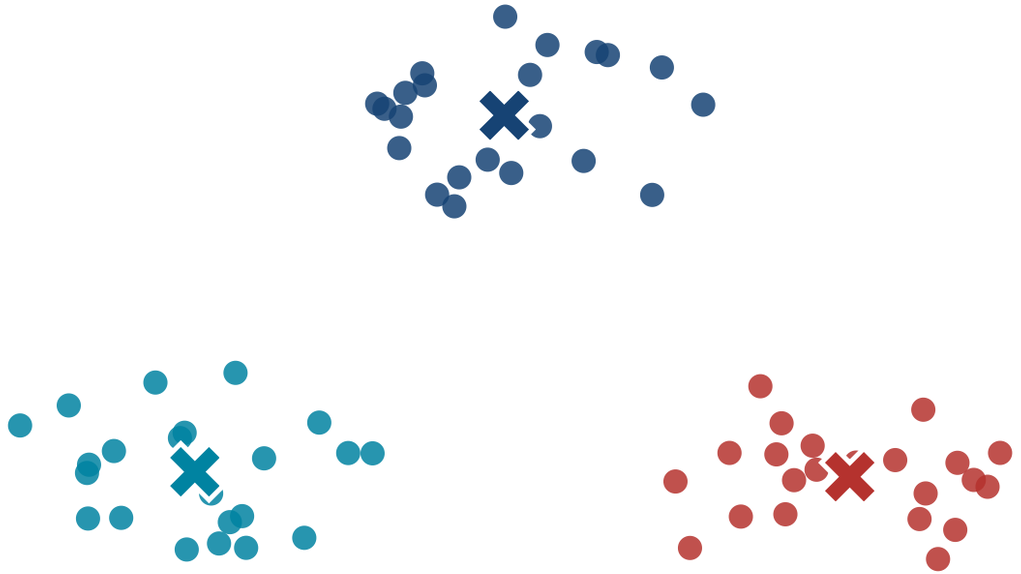
Hard Assignments (Recap)

Guiding Question: k-means gives every point exactly one cluster. What is lost for a point that sits between two?

k-Means in One Slide

Minimise $\mathrm{SSE} = \sum_j \sum_{\mathbf{x} \in C_j} \|\mathbf{x} - \boldsymbol{\mu}_j\|^2$ by alternating: **assign** each point to the nearest centroid, **move** each centroid to the mean of its points. SSE never increases — it stops in a local optimum that depends on the start.

Move each centroid to the mean of its points (round 3)



Speaker notes

Recap, not teaching: every student has run k-means. Odd clicks assign, even clicks move; three rounds. Two things to plant for Module 2. First, the loop alternates between "which cluster?" and "where is the cluster?" — EM will alternate between the same two questions. Second, each assignment is a hard yes/no: a point is 100% in one cluster even when it sits on the boundary.

Where k-Means Fails

failure	what k-means assumes	fixed by a Gaussian mixture?
clusters of different spread	every cluster equally wide (distance only)	✓ each component has its own variance
elongated clusters	round clusters	✓ each component has its own covariance
clusters of different size	equal shares: the boundary lies halfway between centroids	✓ mixing weights π_k
a point between two clusters	one label per point	✓ a probability per cluster
rings, spirals	one blob per cluster	✗ still blobs — needs another representation
how many clusters	k is given	✗ still chosen — but by held-out likelihood

Speaker notes

A table: six failure modes compared on the same two columns. The point of the last column: a Gaussian mixture is not a better k-means across the board. It fixes everything that comes from "distance only, one label", and fixes nothing that comes from "clusters are blobs". Adapted from the Data Science unit's four failure modes (exemplar-based-failure-examples).

A Point Between Two Clusters

A point sits exactly halfway between two centroids. k-means must answer "cluster 1" or "cluster 2". What **should** an algorithm say?

> k-means minimises squared distance with **hard assignments**: fast, simple, and unable to say "I am not sure". The next module makes the label a latent variable — and k-means reappears as the limit where the doubt is removed.

Speaker notes

The Module 1 guiding question, now on the table. Let two or three answers come before the click; someone usually suggests "a probability" or "both". Stress that the lost information matters for decisions: an ambiguous case routed to a human needs to be *known* as ambiguous.

A Point Between Two Clusters

A point sits exactly halfway between two centroids. k-means must answer "cluster 1" or "cluster 2". What **should** an algorithm say?

"About half and half" — which needs the cluster label to be a **random variable** with a distribution, not a value.

> k-means minimises squared distance with **hard assignments**: fast, simple, and unable to say "I am not sure". The next module makes the label a latent variable — and k-means reappears as the limit where the doubt is removed.

Module 2

Mixture Models and EM

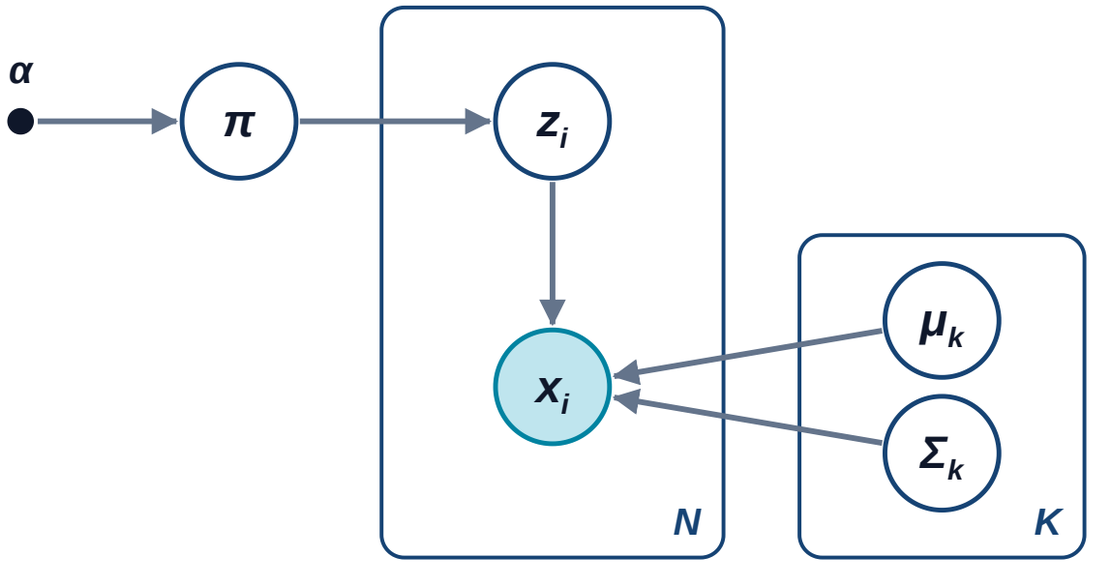
Guiding Question: If the cluster label were just another unobserved variable, could we infer it the way we infer anything else — and what would still be a guess?

The Label as a Hidden Variable

Thirty flats, described by their **price per m²**.
The listing does not say which district a flat is in — suburbs or old town.
A **Gaussian mixture** tells how such data arise:

1. pick a component $z \in \{1, \dots, K\}$ with probability π_z,
2. draw $x \sim \mathcal{N}(\mu_z, \sigma_z^2)$.

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \sigma_k^2)$$



The plate of Lecture 05: z_i is **latent** — never observed.

Speaker notes

Model the joint, including what is hidden: the first of Lecture 01's three moves, done on purpose. The sum is marginalisation over z (Lecture 02). One dimension keeps the formulas readable; in d dimensions σ^2 becomes a covariance matrix Σ, exactly as in the plate. The district is a real latent variable: every listing has one, the data set just does not record it.

Why There Is No Closed Form

For one Gaussian, maximum likelihood is the sample mean and variance (Lecture 03). For a mixture, the log-likelihood is a **log of a sum**:

$$\ell(\theta) = \sum_{i=1}^{N} \log \sum_{k=1}^{K} \pi_k \mathcal{N}(x_i \mid \mu_k, \sigma_k^2)$$

The sum inside the log ties all components together; setting the gradient to zero gives no formula.

If you **knew** each flat's district, how would you fit the two Gaussians? If you **knew** the two Gaussians, how would you find each flat's district?

Speaker notes

Math first (P5): the log of a sum is the whole difficulty. With labels, the log would sit directly on each $\pi_k N(\cdot)$ and split into separate, easy problems — that is why the two questions on the slide each have an easy answer. EM uses the second answer to fill in the first.

Why There Is No Closed Form

For one Gaussian, maximum likelihood is the sample mean and variance (Lecture 03). For a mixture, the log-likelihood is a **log of a sum**:

$$\ell(\theta) = \sum_{i=1}^{N} \log \sum_{k=1}^{K} \pi_k \mathcal{N}(x_i \mid \mu_k, \sigma_k^2)$$

The sum inside the log ties all components together; setting the gradient to zero gives no formula.

If you **knew** each flat's district, how would you fit the two Gaussians? If you **knew** the two Gaussians, how would you find each flat's district?

Known labels: count, average and take the spread per district — maximum likelihood per group. Known Gaussians: Bayes' theorem per flat. **EM alternates the two.**

The E-Step: Bayes for Every Point

Given the current parameters, the **responsibility** of component k for point i is the posterior of the hidden label:

$$\gamma_{ik} = P(z_i = k \mid x_i) = \frac{\pi_k \cdot \mathcal{N}(x_i \mid \mu_k, \sigma_k^2)}{\sum_j \pi_j \mathcal{N}(x_i \mid \mu_j, \sigma_j^2)}$$

Prior (the share of component k) times likelihood (how well k explains x_i), normalised over the components.

> The same computation as Lecture 02's disease test — once per data point. A flat halfway between the two districts gets about $0.5/0.5$.

Speaker notes

Say the E-step out loud as Bayes' theorem. This is the part of EM that performs the ideal: the hidden label is conditioned on the data and kept as a whole distribution, not collapsed to a guess.

The M-Step: Maximum Likelihood With Soft Counts

$$N_k = \sum_i \gamma_{ik}, \qquad \pi_k = \frac{N_k}{N}$$

$$\mu_k = \frac{1}{N_k} \sum_i \gamma_{ik} x_i, \qquad \sigma_k^2 = \frac{1}{N_k} \sum_i \gamma_{ik} (x_i - \mu_k)^2$$

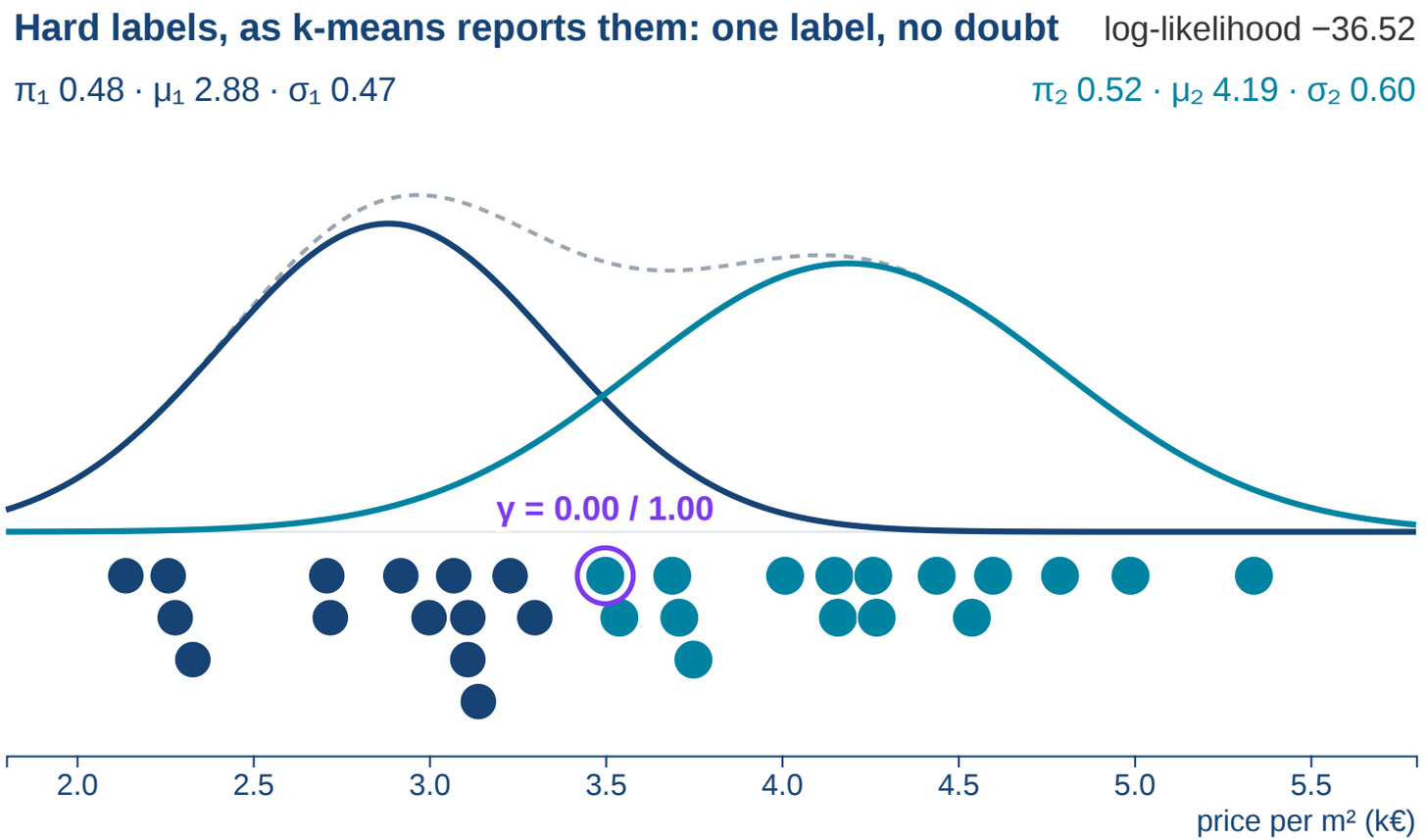
Expectation–Maximisation

1. Start with a guess for π_k, μ_k, σ_k.
2. **E-step:** compute every γ_{ik} (Bayes).
3. **M-step:** refit π_k, μ_k, σ_k — each point counts γ_{ik} times.
4. Repeat until $\ell(\theta)$ stops rising — it never decreases.

Speaker notes

The M-step is Lecture 03's maximum likelihood (share, mean, variance), with each point counted fractionally. With $\gamma \in \{0, 1\}$ it would be exactly "fit each group separately". The monotone likelihood is EM's guarantee; like k-means, it reaches a local optimum that depends on the start.

Watching EM



Predict before the first click: which flats will come out half-coloured — and does the flat at 3.5 k€/m² end up decided?

Speaker notes

Stepped viz (P6). Start: two Gaussians placed badly (μ = 2.2 and 3.4). Click 1: E-step — colour is responsibility (navy = component 1, teal = component 2). Click 2: M-step — the curves jump towards the data; the log-likelihood rises from −51.6 to −36.9. Clicks 3–4: a second round, small changes. Click 5: after 60 iterations — π ≈ 0.48/0.52, μ ≈ 2.89 and 4.19, log-likelihood −36.52; the ringed flat at 3.5 has γ = 0.49/0.51. Click 6: the same flat with a hard label, as k-means would report it. Ask what was lost between clicks 5 and 6.

EM and k-Means

	k-means	EM on a Gaussian mixture
assignment	hard: one cluster per point	soft: responsibilities γ_{ik}
cluster shape	equal and round (distance only)	own variance (covariance) and weight
objective	SSE	log-likelihood $\ell(\theta)$
output	labels and centroids	a density $p(x)$ and a posterior per point

k-means is the limit of EM with equal, spherical components whose variance shrinks to zero: the responsibilities collapse to 0 or 1, and the M-step becomes "move to the mean".

What EM Still Drops

✓ **The hidden labels:** a full posterior per point — the ideal, carried out.

✗ **The parameters:** π_k, μ_k, σ_k are **one point estimate** — Lecture 03's peak. The plate draws them as unknowns; EM gives them no error bar.

✗ **A peak that may not exist:** a component that shrinks onto one point has $\sigma_k \to 0$ and infinite likelihood. In practice: a floor on σ_k, or a prior (MAP).

✗ **The number of components:** the training likelihood always rises with K. Choose K by the **held-out** log-likelihood (Lecture 06), not by an elbow.

> EM is a **partial shortcut**: Bayes for what is hidden, a peak for the parameters. Keeping a posterior over π, μ, σ as well needs sampling or variational inference.

Speaker notes

This slide places EM honestly on the course arc. Lecture 01's table called Lecture 07 "no shortcut"; for EM that is true of the labels only. The collapse is a real failure, not a curiosity: it is why every GMM library has a regularisation parameter on the covariance. The held-out log-likelihood is the log-loss of Lecture 06, now for a density.

Mini-Exercise

Component 1 has weight $\pi_1 = 0.75$, component 2 has $\pi_2 = 0.25$. For one flat, $\mathcal{N}(x \mid \mu_1, \sigma_1^2) = 0.2$ and $\mathcal{N}(x \mid \mu_2, \sigma_2^2) = 0.6$.

What are its responsibilities?

Speaker notes

Head arithmetic, one question. The transferable point: responsibilities are posteriors, so prior and likelihood both count — exactly Lecture 02's base-rate lesson, for a cluster label.

Mini-Exercise

Component 1 has weight $\pi_1 = 0.75$, component 2 has $\pi_2 = 0.25$. For one flat, $\mathcal{N}(x \mid \mu_1, \sigma_1^2) = 0.2$ and $\mathcal{N}(x \mid \mu_2, \sigma_2^2) = 0.6$.

What are its responsibilities?

$0.75 \cdot 0.2 = 0.15$ and $0.25 \cdot 0.6 = 0.15$, so $\gamma = 0.5/0.5$. The prior favours component 1 three to one, the likelihood favours component 2 three to one — they cancel. Nearest-centroid assignment would ignore the weights.

Takeaway

A mixture model makes the cluster label a **latent variable with a posterior**. EM alternates **Bayes for the labels** (E) and **weighted maximum likelihood for the parameters** (M): the posterior over what is hidden, a peak for the rest. k-means is what remains when the doubt is squeezed out.

Module 3

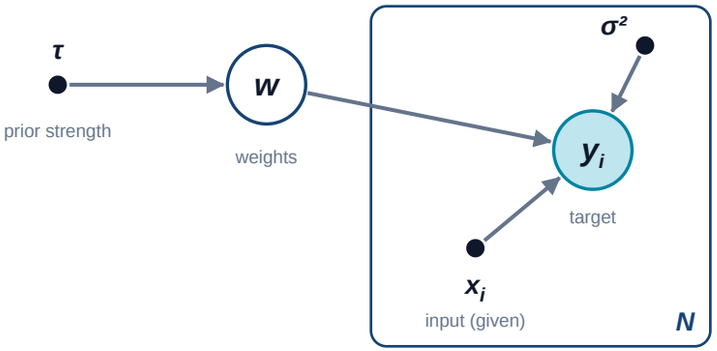
Keeping the Uncertainty

Guiding Question: Your model prices a 600 m² villa at €2.1M — five times larger than any flat it was trained on. Should it say how sure it is, and how much of its doubt could data remove?

Keeping the Distribution

Lecture 01's house-price model: **20 flats of 40–120 m²**, and a question about a **600 m² villa**. The model and the plate of Lecture 05:

$$y_i = w_0 + w_1 x_i + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$



- → **Likelihood:** the Gaussian noise model of Lecture 04, $\sigma = 40$ k€.
- → **Prior:** $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{S}_0)$ — intercept within about ±1,000 k€, slope within about ±20 k€/m². Weak.
- → **Posterior:** Gaussian again — the prior is **conjugate**.

Speaker notes

Maximum likelihood (Lectures 03–04) kept the peak of p(w | D); MAP kept the peak of a posterior. Now the whole posterior is kept — Lecture 02's Beta posterior, for two parameters. S_0 = diag(500², 10²): ±2 prior sd is ±1,000 k€ for the intercept and ±20 k€/m² for the slope — generous enough that the data decide. σ is treated as known to keep the posterior Gaussian; with σ unknown the predictive distribution becomes a Student t, slightly wider.

Precision Adds Up

$$\mathbf{S}_N^{-1} = \mathbf{S}_0^{-1} + \frac{1}{\sigma^2}\mathbf{X}^\top\mathbf{X}, \qquad \mathbf{m}_N = \frac{1}{\sigma^2}\mathbf{S}_N\mathbf{X}^\top\mathbf{y}$$

The update is adding

Posterior precision = prior precision + data precision — Lecture 02's **adding counts**, for Gaussians. Every flat adds $\frac{1}{\sigma^2}\mathbf{x}_i\mathbf{x}_i^\top$.

The centre is the ridge estimate

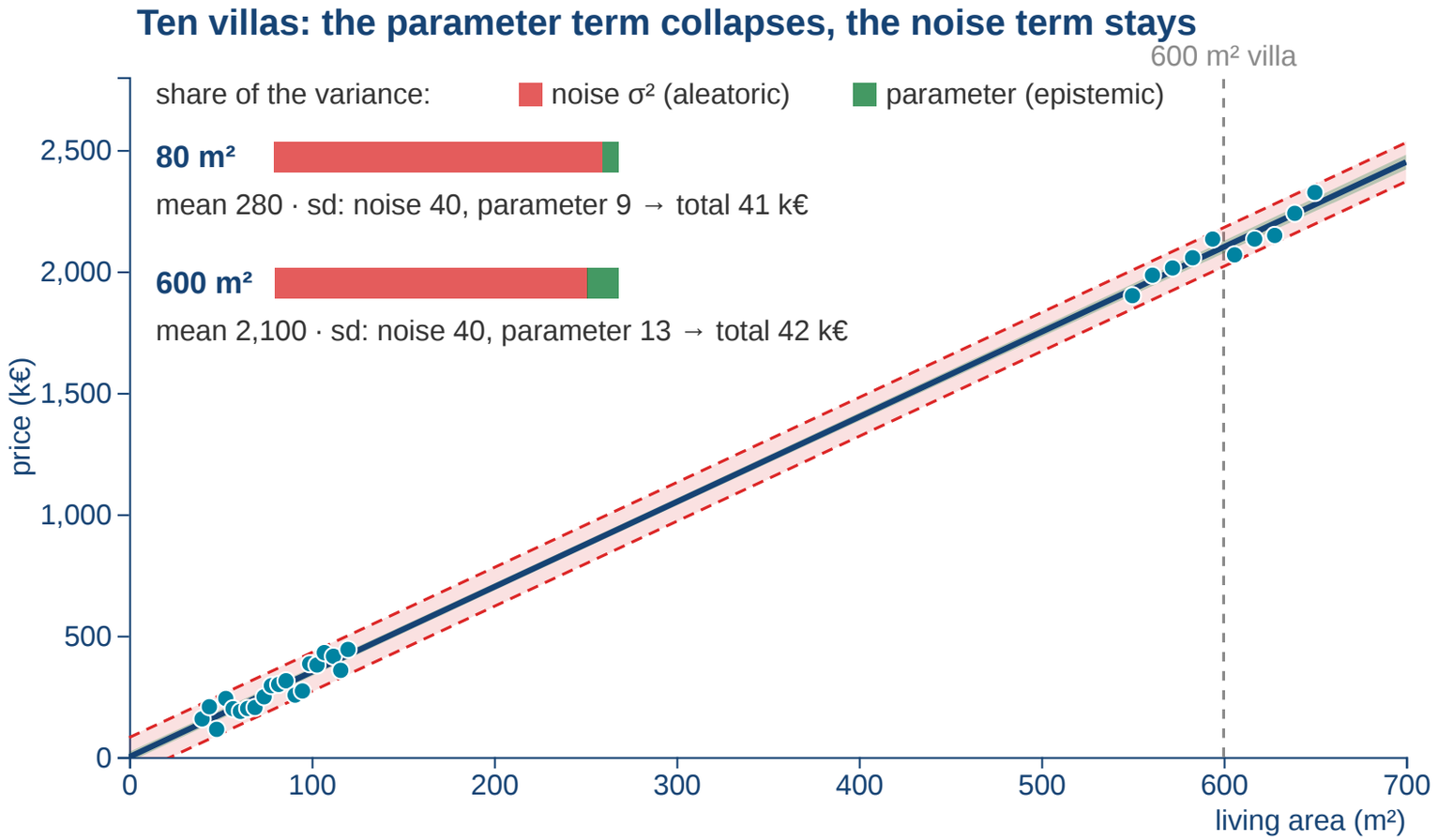
$\mathbf{m}_N$ is the MAP estimate — ridge regression. With this weak prior it is almost least squares: slope **3.496** vs **3.501** k€/m².

What is new is not $\mathbf{m}_N$ but $\mathbf{S}_N$: the slope is 3.50 ± 0.37 k€/m² (one sd) — and that uncertainty is multiplied by 600 m² at the villa.

Speaker notes

Math first, then the reading. The first card is the Gaussian version of Beta(a + h, b + n − h): precisions add like counts. The second card answers the usual question "so Bayesian regression gives a different line?" — hardly: the centre is ridge, and a weak prior barely moves it. What changes is that the posterior also says how far every other line is from implausible. Intercept and slope are strongly negatively correlated (−0.96): lines pivot around the centre of the data, which the next slide shows.

Watching the Band Form



Predict before the last click: ten villas of 550–650 m² join the data. Which part of the band shrinks — and which part cannot?

Speaker notes

Stepped viz (P6). On entry: the 20 flats and the question at 600 m². Click 1: keep the peak (Lecture 04) — least squares with the plug-in band ±2σ = ±80 k€, the same width everywhere. Click 2: fifteen lines drawn from the posterior — they agree on the flats and fan out at 600 m². Click 3: the predictive band (±2 sd) split into its two parts; the bars show the shares of the variance: at 80 m² the noise is 95%, at 600 m² the parameter term is 96%. Click 4: ten villas — the parameter part collapses (at 600 m²: 191 → 13 k€), the noise part does not move. The dashed red lines are ±2σ around the mean: the noise alone.

Two Terms of One Prediction

$$p(y^* \mid x^*, D) = \mathcal{N}\left(\mathbf{m}_N^\top \mathbf{x}^*,\ \ \sigma^2 + \mathbf{x}^{*\top} \mathbf{S}_N \mathbf{x}^*\right)$$

■ noise term — aleatoric, fixed · ■ parameter term — epistemic, shrinks with data

query	mean	noise sd	parameter sd	total sd
80 m² (inside the data)	280 k€	40	9	**41**
300 m²	1,049 k€	40	81	**90**
600 m² — the villa	2,098 k€	40	191	**195**
600 m², after ten villas	2,100 k€	40	13	**42**

> Near the data the **noise** dominates, at the villa the **ignorance**. Ten villas remove the ignorance; the noise stays — Lecture 01's two examples.

Speaker notes

The promise of Lecture 01 (sources-of-uncertainty: "a Bayesian one can report them separately"), paid with numbers. Variances add, not standard deviations: at 600 m², $40^2 + 191^2 = 1{,}600 + 36{,}419 \approx 195^2$. Hence a small term vanishes next to a large one — 9 k€ of parameter sd next to 40 k€ of noise changes the total by 1 k€. At 120 m², the edge of the data: parameter sd 17, total 44 k€. All numbers from σ = 40 k€ known and $S_0 = \mathrm{diag}(500^2, 10^2)$; the flat-prior limit gives the same to one digit. A thousand more flats of 40–120 m² would shrink the parameter term at 600 m² only slowly, because it is driven by the distance from the data; the noise term would not move at all.

What the Posterior Cannot Know

The band is uncertainty **within the linear model**. Suppose villas actually sell at 5,000 €/m²: the villa is worth **€3.0M** — far outside 2,098 ± 2 · 195 k€, i.e. €1.71M to €2.49M.

The world is noisy

shows up in σ^2, the noise term — accept it, or measure new attributes

Too few data here

shows up in $\mathbf{x}^{*\top}\mathbf{S}_N\mathbf{x}^*$, the parameter term — collect data there

The family is wrong

shows up **nowhere** in the formula — check on held-out data where you predict; compare model families

A wide band is an honest "I don't know". A narrow band is not a proof — it can be confidently wrong if the straight line itself is wrong.

Speaker notes

This is the third branch of Lecture 01's reasoning cycle ("noise, too little data, or the wrong model?"), now with the formula in hand: two of the three answers have a term, the third has none. Bayesian regression measures ignorance about w; it is silent about ignorance of the model. Only data at 600 m² — or a richer model compared by held-out likelihood — can reveal it.

Deciding With the Predictive Distribution

A lender lends up to the value it is 97.5% sure the property exceeds: the mean minus two standard deviations.

€198k

an 80 m² flat: $280 - 2 \cdot 41$

€1.71M

the villa today: $2{,}098 - 2 \cdot 195$

€2.02M

after ten villa sales: $2{,}100 - 2 \cdot 42$

Ten comparable sales are worth about **€0.3M of credit**: buying data pays where the **parameter** term dominates. Near the data it buys nothing — the €40k noise stays.

Speaker notes

Lecture 01's roadmap promised "keep uncertainty all the way to the decision". This is the simplest version: the decision reads a quantile of the predictive distribution, so both terms enter it. The same logic tells the lender where information is worth paying for: at 600 m², a few comparable sales change the decision by €0.3M; at 80 m², a thousand more flats would change it by almost nothing. A point estimate would lend €2.1M on the villa and be exposed to the whole €195k spread without knowing it.

Mini-Exercise

Model A (maximum likelihood with the plug-in noise of Lecture 04) and model B (Bayesian) both predict about **€2.1M** for the 600 m² villa. A reports **±80 k€**.

What does B report — and which one do you show the lender?

Speaker notes

Two minutes; the numbers are on the previous slides. Two precise remarks for the discussion. (1) A's ±80 k€ is the plug-in interval $N(\hat{w}^T x, \sigma^2)$ of Lecture 04. The textbook frequentist prediction interval, $s \cdot t \cdot \sqrt{(1 + x^T (X^T X)^{-1} x)}$, also widens with the distance from the data — here ±410 k€ with the t quantile. The parameter term is not a Bayesian privilege; reading it as a probability about this villa is. (2) B's centre is the ridge estimate, practically least squares with this weak prior.

Mini-Exercise

Model A (maximum likelihood with the plug-in noise of Lecture 04) and model B (Bayesian) both predict about **€2.1M** for the 600 m² villa. A reports **±80 k€**.

What does B report — and which one do you show the lender?

B reports about **±390 k€** ($2 \cdot 195$): its parameter term is large this far from the data. Show **B** — not because its centre is better (both are ≈ €2.1M) but because A claims the precision of an 80 m² flat for a house it has never seen anything like.

Takeaway

Bayesian regression keeps a **posterior over the weights** and predicts with a **predictive distribution** whose variance has two terms: the **noise** term (aleatoric) that no data removes, and the **parameter** term (epistemic) that grows off the data and shrinks where data is collected. What neither term covers is a wrong model family.

Module 4

Closing the Arc

Guiding Question: Seven lectures, one ideal, several shortcuts. Which questions can each method you know still answer — and what would it cost to answer the rest?

The Shortcuts, Revisited

Lecture 01's table, with what the course has learned about each row:

shortcut	what it dropped	bought back by	at the cost of
Keep the peak · 03–04	how sure we are about the parameters	the posterior (Module 3)	a prior, an integral
Assume it apart · 03	the dependences assumed away	a graph that draws them	more parameters
Only $p(y \mid x)$ · 04	every question about x	a generative model	modelling x
Graph the joint · 05	the edges not drawn	more edges	costlier inference
Keep the loss · 06	every probability	calibration, log-loss, error bars	held-out data
EM, partial · 07	how sure we are about π, μ, σ	priors, sampling or VI	computation
No shortcut · 07	nothing — within the model family	—	a prior; the family itself

> Every shortcut is a trade of an answer for computation or data. Knowing which one a method takes tells you which question it can no longer answer.

Speaker notes

Close the arc with the table the course opened with (L01 roadmap), extended by two columns. Walk it top to bottom quickly — each row is one lecture's takeaway. Two rows need a sentence: EM is only partly the ideal (Module 2), and even Bayesian regression is exact only within its model family (Module 3, "what the posterior cannot know"). The last column is the honest reason why the shortcuts exist: the ideal is expensive.

What Comes After

- → **Gaussian processes** — Bayesian regression with infinitely many basis functions: the two-term band of Module 3, without the straight line.
- → **Bayesian neural networks and deep ensembles** — the parameter term for models with millions of weights, approximately.
- → **Sampling (MCMC) and variational inference** — conditioning on the data when there is no closed form; EM is the special case that keeps a posterior only for the hidden labels.

Model the joint, condition on the data, read off answers — and when you take a shortcut, **name it and know what it drops**. A prediction worth acting on says how sure it is, and why: noise, ignorance, or a model that may be wrong.

Speaker notes

The outlook Lecture 01 promised ("and what comes after"). One line each; the point is that every one of them is the same programme with a different way of paying for the conditioning step. End on the course's one sentence.